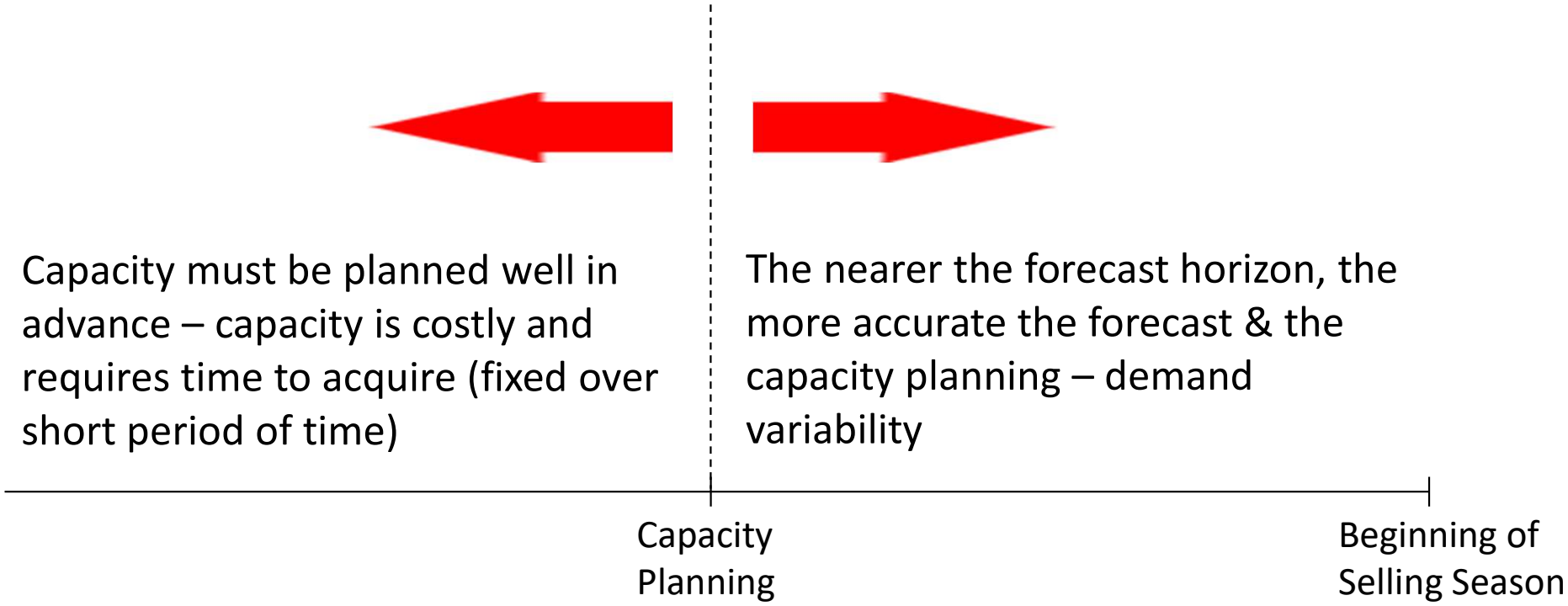
OPIM 201 OPERATIONS MANAGEMENT

CAPACITY PLANNING (NOT TESTED)

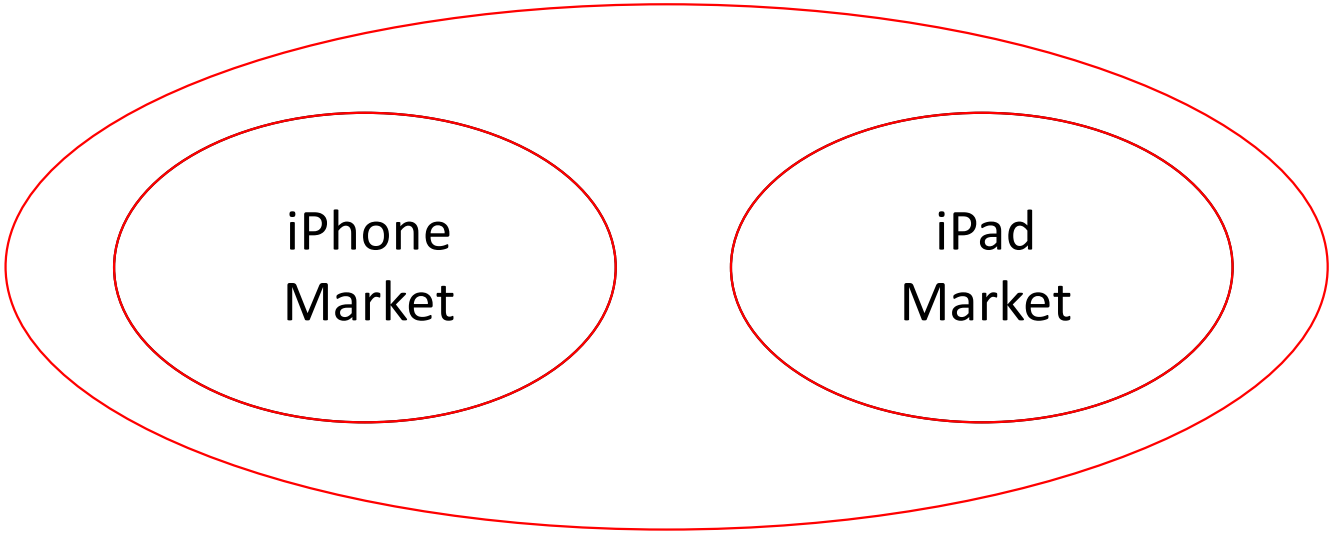
Outline

- Introduction to Capacity Planning
- Understand how capacity planning is done in different phases
  - Aggregate Planning

Capacity Planning Dilemma



Effect of Aggregating Demand



Which forecast is more accurate, individual forecasts for iPhone & iPad, or aggregate forecast of total demand?

	iPhone	iPad	Total
Forecast	100	100	200
Demand	120	80	200
Error	–20	+20	0

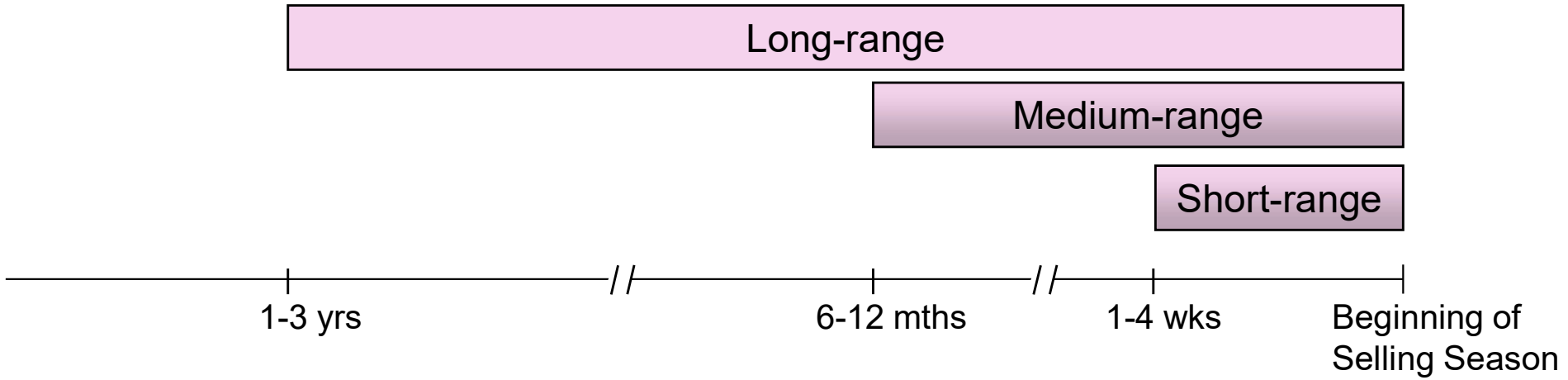
Characteristics of Forecasts

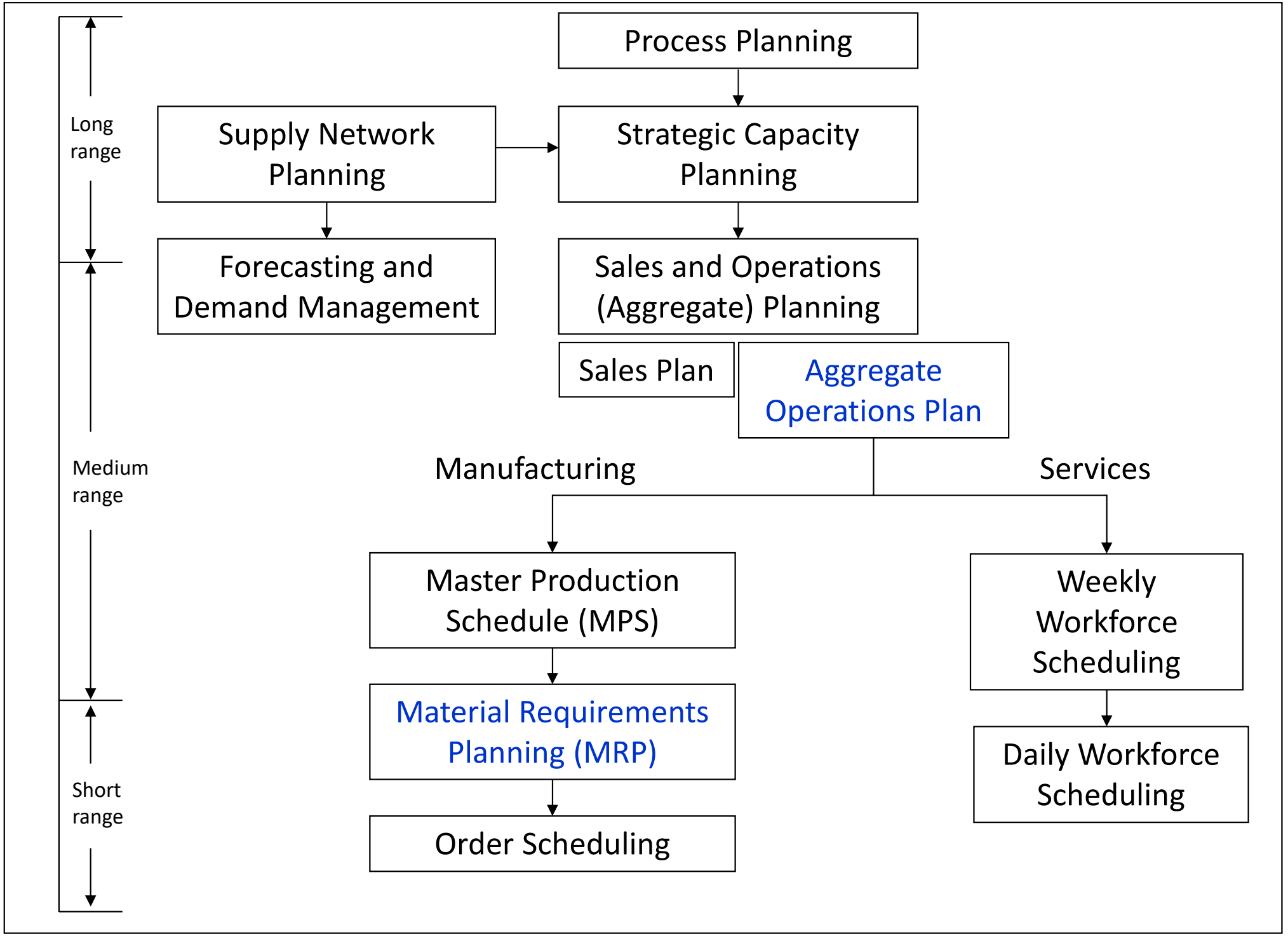
- Forecasts are always wrong
- Long-term forecasts are less accurate than short-term forecasts – different forecast details for different forecast horizons
  - Raw material purchases for the next week
  - Annual electricity generation capacity in Texas for the next 30 years
- Aggregate forecasts are more accurate than disaggregate forecasts
  - Variance of aggregate demand is smaller
    - Demand for meat during Christmas has less variability than the total variability in the demand for chicken, turkey, beef.
  - Several ways to aggregate
    - By product groups
    - By location
    - By time

	iPhone	iPad	Total
Forecast	100	100	200
Demand	120	80	200
Error	–20	+20	0

Three Phases of Capacity Planning

- Strategic Capacity Planning (Long-range)
  - System design decisions (product/process design)
  - Usually performed in annual increments
- Aggregate Planning (Medium-range)
  - Employment, output etc. at aggregate level
  - Usually performed with weekly, monthly or quarterly increments
- Operational Planning (Short-range)
  - Detailed production and scheduling planning, Job assignments etc. (e.g., MRP/ERP)
  - Usually with weekly or daily increments




Process Planning
Long range
Supply Network Planning
Strategic Capacity Planning
Forecasting and Demand Management
Sales and Operations (Aggregate) Planning
Sales Plan
Aggregate Operations Plan
Medium range
Manufacturing
Services
Master Production Schedule (MPS)
Weekly Workforce Scheduling
Material Requirements Planning (MRP)
Daily Workforce Scheduling
Short range
Order Scheduling

Role of Aggregate Planning in a Supply Chain

- Capacity planning and decisions for some resources can be made at an aggregate level (or product family) instead of stock keeping unit (SKU) level (or individual product) when forecast is less accurate
  - Examples: production rate, workforce, machine capacity, overtime, subcontracting, inventory, backlog/stockouts, and pricing
- Aggregation reduces variability
  - Detailed forecasts at SKU level carry a lot of uncertainty & are not reliable for intermediate range planning
- Aggregation offers flexibility
  - If there is variability why bother making detailed plans, inputs will change anyway; instead make plans that carry a lot of flexibility
- Manufacturing firm aggregate plan: Production plan;
  Service firm aggregate plan: Staffing plan

Example: Suntory Beverage & Food Limited



Example: Suntory Beverage & Food Limited (cont'd)

A Suntory bottling factory processes three different types of Whisky: Scotch Whisky, Japanese Whisky & Irish Whiskey. While the bottling process is highly automated, there are labor needs for activities including inserting corks/screwcaps, quality control, labeling and packaging. An aggregate plan for planning resources including(e.g. workforce) at an aggregate level can be built based on an aggregate forecast:

Daily Forecast	Mar	Apr	May	Jun	Jul	Aug
Scotch Whisky	24600	24300	25500	15000	16500	17400
Japanese Whisky	10800	12600	11400	9000	8610	9600
Irish Whiskey	8400	9000	10500	7500	7200	9000
Aggregate Forecast	43800	45900	47400	31500	32310	36000

The aggregate forecast will be again disaggregated into detailed individual forecast nearer to actual production when the individual forecast can be made more accurately.